BHUSHAN WALUNJ

8446533962 | bhushanwalunj2718@gmail.com | linkedin.com/in/bhushan-walunj | github.com/bhushan191004 | bhushan191004.github.io | Pune, Maharashtra

PROFESSIONAL SUMMARY

Final-year B.E. (Electronics & Telecommunication) student and Computer Vision / Generative AI engineer. Trained and deployed a YOLOv5 detection system, built a RAG chatbot and a LangGraph multi-agent app, and completed three internships in Python, AI and automation. Seeking AI/ML Engineer and Python AI Developer roles.

TECHNICAL SKILLS

AI / ML: PyTorch · YOLOv5 · YOLOv8 · OpenCV · Hugging Face · Deep Learning · Model Training & Evaluation
Generative AI: LLMs · RAG · LangChain · LangGraph · ChromaDB · FAISS · Prompt Engineering · AI Agents
Backend & Deployment: Python · Flask · FastAPI · Streamlit · REST APIs · SQLite · Git
Also: n8n · AutomationEdge (RPA) · C++ · NumPy · Pandas

PROJECTS

AI-Based Helmet Violation Detection System (B-DESK) **2024 – 2025**
Python · YOLOv5 · PyTorch · Flask · OpenCV

- Trained a custom YOLOv5 helmet detector with tuned anchors and data augmentation, reaching ~0.83 mAP@50.
- Deployed as a Flask app supporting image upload and live webcam detection through an optimised OpenCV inference pipeline.

LoanDesk — RAG-Based Loan Advisory Chatbot **2025**
LangChain · ChromaDB · Python · Streamlit · LLM API

- Built a RAG pipeline over loan-policy PDFs: document ingestion, chunking, embeddings, ChromaDB retrieval and LLM answer generation, with sub-2s response latency.
- Built a Streamlit interface for PDF upload and conversational querying.

CityFix AI — Multi-Agent Civic Issue Reporting System **2026 – Present**
LangGraph · Groq (Llama 3.3 70B) · Streamlit · SQLite · Plotly

- Designed a multi-agent LangGraph workflow that identifies civic issues from user photos (potholes, broken streetlights), categorises them and routes them automatically.
- Added location pinning on a map and auto-generated PDF reports, removing manual reporting work.

EXPERIENCE

Generative AI & Agentic Systems Intern · Lenovo LEAP × AICTE (Remote) **June 2026 – July 2026**

- Developed LLM-based applications and agentic AI pipelines; applied prompt engineering.
- Delivered CityFix AI, a multi-agent system, as the capstone project.

RPA Trainee Intern · ValueDX Technologies, Pune **June 2026 – Sep 2026**

- Built RPA workflows in AutomationEdge Process Studio using JSON, REST APIs and document processing.
- Implemented exception handling and debugging, and deployed bots.

Python Developer Intern · MASS IT Solutions LLP, Pune **Jan 2026 – Feb 2026**

- Built computer vision modules in Python, PyTorch and YOLOv5, covering the pipeline from preprocessing and augmentation to evaluation and inference.
- Tuned model performance and reduced inference latency for real-time use.

EDUCATION

B.E. — Electronics & Telecommunication · SKNCOE, Pune **Aug 2023 – Present**
CGPA: 7.63 | Latest SGPA: 8.52

HSC (Science) · GRPS College of Science, Pune · Maharashtra State Board **2021 – 2023**
Percentage: 63.5%

SSC · Maharashtra State Board **Percentage: 94%**

CERTIFICATIONS

Gen AI 101 · FutureSkills Prime, Nasscom (Apr 2026) | Gen AI Engineering Mastermind · Outskill (2026) | AutomationEdge Foundation & Intermediate · AutomationEdge Academy (2026)